

IIT Kharagpur

Department of Mathematics

Semester-5 (DC)

Statistics Software Lab

(LTP- 0-0-3,CRD- 3)

Prerequisites(If any):

Probability & Statistics (MA20205), Statistical Inference(MA41108)

Content:

1. Introduction to R / Python [3 Lectures]
2. Random number generation (Discrete) [3 Lectures]
3. Random number generation (Continuous) [3 Lectures]
4. Random number generation (Joint) [3 Lectures]
5. Study of sampling distributions / transformations [6 Lectures]
6. WLLN and CLT [3 Lectures]
7. Estimation (MLE, MME) [3 Lectures]
8. Empirical Fisher Information [3 Lectures]
9. Testing of hypothesis (One Sample) [3 Lectures]
10. Testing of hypothesis (Two Samples) [6 Lectures]
11. Testing of hypothesis (Dependent distributions) [3 Lectures]
12. Goodness-of-fit test [3 Lectures]

References:

1. Kerns, G. J. (2010). Introduction to probability and statistics using R. [Text]
2. Venables, W. N., Smith, D. M., & R Development Core Team. (2009). An introduction to R.[Text]
3. Peter, D. (2008). Introductory statistics with R. *Springer, New York Polge H (1970)*
4. Albert, J., & Rizzo, M. (2012). R by Example. Springer Science & Business Media.

Instructors: S Kumar , N Gupta, S Khare , B Banerjee, A Banerjee